

1: Introduction to WAN Technologies
Theory :WAN vs. LAN: Key differences and use cases.
MPLS basics.
Practical :Observe MPLS routing in a pre-configured network.
Task: Document MPLS benefits over traditional routing.

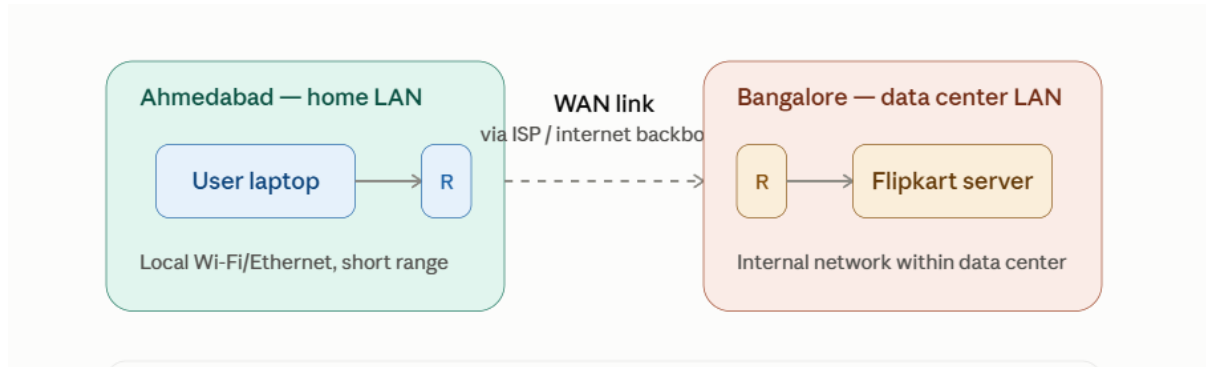
Task 1.

Create a comparison table listing at least 3 key differences between WAN and LAN, and give one real-world example of each from apps or services you use daily (for example, IRCTC, Instagram, or Zomato).

Feature	LAN (Local Area Network)	WAN (Wide Area Network)
Coverage	Covers a small area such as a home, office, or building	Covers large geographical areas such as cities, states, or countries
Speed	Generally provides higher speed and lower latency	Usually has higher latency compared with LAN
Ownership	Usually owned and managed by an individual or organization	Often uses networks/services provided by telecom or Internet providers
Example	A company office network connecting PCs, printers and servers	Connecting an Ahmedabad office to a data center in Bangalore
Daily-life example	Zomato/Flipkart LAN inside a company's office or data center	IRCTC/Flipkart users accessing servers located in another city

Task 2.

Draw a simple network diagram using any drawing tool (or on paper and upload a photo) that shows how a user in Ahmedabad could connect to a Flipkart data center in Bangalore using a WAN, labeling the LAN and WAN segments clearly.



The path:

your laptop joins the Ahmedabad LAN (home Wi-Fi/router) → traffic exits over the WAN (your ISP's network, then the wider internet backbone) → arrives at Flipkart's Bangalore data center LAN, which routes it internally to the actual server.

Task 3.

Research and write a short summary (5-7 lines) explaining what MPLS is and how it improves data routing compared to traditional IP routing.

- MPLS (Multiprotocol Label Switching) is a technology used to forward network traffic using short labels instead of examining the full IP routing information at every router.
- When a packet enters an MPLS network, a label is attached to it. MPLS routers use this label to quickly determine where the packet should go.
- This creates a predefined path called a Label Switched Path (LSP).
- MPLS can provide better traffic management, predictable performance, and Quality of Service (QoS).

- It can also help companies efficiently prioritize important applications such as voice, video, and business traffic.
- Therefore, MPLS provides more controlled traffic forwarding than traditional destination-based IP routing.

Task 4.

Observe the pre-configured MPLS network in the provided Packet Tracer or GNS3 file, and note down two routing behaviors that are different from standard IP routing.

Hint: Look for how labels are used and how packets move between routers.

Behaviour 1 — Label-based forwarding

Traditional IP routing generally examines the **destination IP address** and consults the routing table.

In MPLS:

Incoming packet

↓

MPLS label

↓

Label lookup

↓

Next router

The router can forward traffic based on the MPLS label rather than performing a normal IP routing-table lookup for every hop.

Behavior 2 — Label Switched Path (LSP)

MPLS traffic can follow a predetermined **Label Switched Path** through the MPLS network. Intermediate routers use labels to forward the traffic along that path.

After you upload the Packet Tracer/GNS3 file, I can give you the exact observations in assignment-ready wording.

Task 5.

List three specific benefits MPLS offers over traditional routing for companies running high-traffic apps like YouTube or BookMyShow, and briefly explain how each benefit helps in real-world scenarios.

Benefit	Explanation	Real-world example
1. Traffic Engineering	MPLS allows companies to control how traffic travels through the network and avoid congested links.	YouTube: Heavy video traffic can be distributed through suitable network paths to reduce congestion.
2. Quality of Service (QoS)	Important traffic can receive higher priority than less-critical traffic.	BookMyShow: During a major movie release, transaction/payment traffic can be prioritized to improve reliability.
3. Better Reliability & Predictable Performance	MPLS can use controlled paths and backup routes, helping maintain connectivity when a network link fails.	BookMyShow/large businesses: During high-demand periods, resilient WAN connectivity helps keep websites and applications available.